

Quarterly Business Review

PayFlow Inc. — Q1 2026 | Saravanan Ramesh, Sr. Technical Account Manager

Target SLA	~65ms	\$36/mo	0
99.95% uptime	Avg response time	Monthly infra cost	P1 incidents

Executive Summary

This quarter PayFlow moved from a single-server deployment to a scalable, self-healing Kubernetes architecture on DigitalOcean Kubernetes Service (DOKS). The deployment demonstrates production-ready patterns — horizontal pod autoscaling, cluster autoscaling, pod anti-affinity HA, and PostgreSQL persistence. Load testing validated automatic scaling from 2 to 3 pods and cluster autoscaling provisioning a new node in ~56 seconds — giving PayFlow the infrastructure foundation to support its next growth phase at \$36/month baseline, 4× cheaper than equivalent AWS EKS architecture.

Current Infrastructure & Performance

Component	Configuration	Cost
DOKS control plane	Managed, NYC1	\$0
Worker nodes (2×)	s-1vcpu-2gb, autoscaler min 2 / max 3	\$24
DO Load Balancer	Standard, Layer 4	\$12
Container Registry	Starter tier, NYC3	\$0
PostgreSQL + PVC	postgres:15-alpine, 1Gi Block Storage	~\$0.10
Total baseline		\$36/mo

Metric	Result	Metric	Result
Avg response time	~65ms	HPA scale-up (2→3 pods)	~60s at 248% CPU
Load test throughput	752 req/sec (50c)	Cluster autoscaler	Node 3 in ~56s
Baseline node CPU	5–6%	Pod recovery time	~22 seconds
vs AWS EKS	4× cheaper (\$36 vs \$159)	P1 incidents	0

Risks & Mitigation

Risk	Severity	Mitigation
No SSL / HTTPS	High	cert-manager + Let's Encrypt — Q2 2026
No user authentication	High	JWT/OAuth2 middleware — Q2 2026
PostgreSQL pod — no HA	High	Upgrade to DO Managed PostgreSQL — Q2 2026
Single-region (NYC1)	High	Active-Passive multi-region NYC1 + SFO3 — Q3 2026
No centralized logging	Medium	Prometheus + Grafana + Loki via Helm — Q3 2026
Manual deployments	Medium	GitHub Actions CI/CD + versioned image tags — Q3 2026
No Layer 7 WAF	Medium	DO LB handles infra DDoS. Add Cloudflare for WAF — Q4 2026

How Containerization Drives Business Outcomes

Performance	Scalability & Reliability	Cost Optimization
• ~65ms avg response time • Load balanced across healthy pods • Readiness probes prevent bad traffic • Rolling deploys with zero downtime • Pod recovery in ~22 seconds • CDN via DO Spaces for static assets	• HPA: 2→3 pods in 60s at 248% CPU • Cluster autoscaler: Node 3 in 56s • Required antiaffinity ensures node HA • Self-healing — failed pods auto-restart • Scales to 1,000+ customers without re-arch	• \$36/mo vs \$159/mo AWS EKS (4× saving) • Free DOKS control plane saves \$72/mo • HPA fills nodes before new Droplets • Single NGINX Ingress LB vs \$12/service • Burst capacity billed hourly, not monthly • Nodepool scale-to-zero cuts idle costs

Recommendations & Roadmap

Quarter	Priority	Recommendation	Cost
Q2 2026	Critical	Managed PostgreSQL + SSL/TLS + User Auth Replace pod DB with DO Managed PostgreSQL. Add cert-manager + Let's Encrypt. Implement JWT/OAuth2 authentication.	+\$15/mo
Q3 2026	High	Multi-region Active-Passive deployment Add SFO3 cluster for geographic HA and West Coast latency. Async DB replication via DO Managed PostgreSQL. DNS failover using DO Load Balancer.	+\$36/mo
Q3 2026	Medium	CI/CD + Observability + DO Spaces CDN GitHub Actions with versioned image tags. RBAC + Network Policies. Prometheus + Grafana + Loki via Helm. DO Spaces (\$5/mo) for invoice PDFs with built-in CDN. PgBouncer connection pooling. Nodepool scale-to-zero for off-peak cost savings.	+\$5/mo
Q4 2026	Low	AI expansion + WAF + Reserved Droplet pricing DO Gradient AI Platform — serverless inference or agent builder for invoice chatbot (no infra to manage). Cloudflare Pro (\$20/mo) for Layer 7 WAF + global CDN. Evaluate 1-year Reserved Droplet pricing once baseline stabilises.	+\$56/mo

High Availability Architecture

	Active-Passive (recommended — Q3 2026)	Active-Active (future — 1,000+ customers)
Traffic	NYC1 serves all; SFO3 on standby	Both regions serve live traffic simultaneously
Failover	Minutes — DNS failover or manual	Instantaneous — automatic LB rerouting
Cost	~\$88/mo (standby runs minimal nodes)	~\$128/mo (both clusters fully provisioned)
DB replication	Asynchronous — simpler to manage	Synchronous — complex, higher latency risk

AI Expansion Roadmap

Phase	Timeline	Description	Cost
Phase 1	Q4 2026	AI invoice chatbot via DO Gradient AI Platform — use Agent Builder with RAG over DO Spaces invoice PDFs, or Serverless Inference for direct LLM calls. No infrastructure to manage. Answers invoice queries, flags overdue accounts, drafts follow-up emails.	Usage-based
Phase 2	Q1 2027	Anomaly detection via Gradient Serverless Inference or Kubernetes CronJob — queries DB nightly, flags unusual payment patterns and churn risk, writes results to alerts table.	~\$0/mo
Phase 3	Future	Dedicated inference on DO GPU infrastructure via Gradient Platform — cost-efficient at high query volume. Upgrade from serverless to dedicated capacity with no re-architecture.	TBD

Account Health	Renewal Risk	Expansion Opportunity
GREEN	Low	High — Managed DB, SSL/Auth, Spaces, Multi-region, AI tooling